

Humberto Gerardo Peña Páez

AI Data Engineer | Lead Data Engineer | Azure & GenAI Specialist

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SUMMARY

Lead AI Data Engineer with 3.5+ years in Azure, specializing in scalable GenAI/LLM pipelines. Trusted to direct a **Data Engineering Factory** and mentor teams to deliver high-quality technical blueprints using Medallion Architecture. Proven expertise in **SAP integrations** and advanced Databricks features like **Lakeflow**. **Databricks Certified Associate (May 2026)** focused on driving operational value through robust engineering.

TECHNICAL SKILLS

Leadership & Architecture: Technical Mentorship, Software Factory Coordination, Architectural Design, Agile Methodologies (Scrum), Project Lifecycle Management.

Data Engineering: Azure Data Factory, Azure Databricks, PySpark, SQL (T-SQL), Medallion Architecture, Delta Live Tables, ETL/ELT, Snowflake.

AI & Development: Generative AI (LLMs), GenAI Agents API, C# Azure Functions, Azure Cognitive Services, Python (Pandas, NumPy, TensorFlow), APIs Integration.

Business Intelligence & Analytics: Power BI (DAX), SQL Server Analysis Services (SSAS/Cubes), SSRS (MDX), SSIS, Data Modeling.

Tools & DevOps: Git, Azure DevOps, CI/CD, Azure Blob Storage, Azure Key Vault, Azure Synapse, Streamlit.

WORK EXPERIENCE

D&C Solutions

September 2025 – Present

Lead Data Engineer & Factory Coordinator

Monterrey, N.L.

- **Technical Leadership & Trust:** Appointed as Lead to coordinate the internal **Data Engineering Factory**, providing architectural guidance and technical mentorship to junior developers to ensure the delivery of scalable, high-quality solutions.
- **Architecture & Standards:** Define and guarantee compliance with the consultancy's **ETL** and **Medallion Architecture** standards, ensuring all data pipelines are built for high performance and long-term maintainability.
- **High-Impact Data Integration:** Oversee the ingestion of complex datasets from **SAP** via **Azure Blob Storage**, guiding the team through the complete transformation process and the application of advanced business rules.
- **Technical Governance:** Act as the primary guardian of data integrity, resolving complex technical roadblocks and performing rigorous code reviews to ensure all deliverables meet enterprise-grade quality standards.

Heineken México

February 2025 – Present

AI Data Engineer

Monterrey, N.L.

- Architected the end-to-end lifecycle of **AI Agents** for telephone platforms using **Azure Functions** and **C#**, developing scalable APIs for real-time conversational intelligence.
- **Architected and deployed** scalable ETL pipelines following the **Medallion Methodology** (Bronze, Silver, Gold) in **Azure Databricks** and **PySpark** to process high-volume data for **Large Language Model (LLM)** consumption.
- **Engineered** an automated "Call Rating" system with **Generative AI**, utilizing **APIs** and **SQL** to extract and evaluate conversation data within a code-controlled **DevOps** environment.
- **Developed and automated** data ingestion workflows in **Azure Data Factory**, ensuring a seamless flow of raw data into AI-ready datasets for strategic company products.
- **Implemented** a personalized "Promotion Recommendation" engine with **Generative AI**, leveraging predictive analysis to improve delivery precision and generate significant operational savings.

Data Engineer / Analytics Developer

May 2024 – February 2025

- **Orchestrated** high-volume, real-time ETL processes using **Azure Databricks** and **PySpark**, implementing the **Medallion Methodology** (Bronze, Silver, Gold) to guarantee data quality and high availability.
- **Designed and automated** end-to-end data pipelines in **Azure Data Factory**, enabling seamless raw data extraction from external servers and efficient incremental storage in **Azure Blob Storage**.
- **Developed** advanced **Power BI** dashboards with complex **DAX** measures and front-end design, providing actionable insights for 4 strategic projects that resulted in significant operational savings.
- **Optimized** real-time processing by automating streaming workflows in **Databricks**, facilitating a **100% success rate** in the transition of projects from development to production.
- **Enhanced** the analytical capacity of the Data Science team by integrating multi-sector data into the **Data Lake** while ensuring strict compliance and code quality in **PySpark**.

BYDSA

May 2023 – May 2024

Data Engineer / BI Developer

Guadalupe, N.L.

- **Engineered** end-to-end data pipelines to ingest and process various file formats into **SQL Server**, establishing a **Medallion-style architecture** across three distinct catalogs (**Raw, Metrics, and Production**).
- **Automated** ETL workflows using **SSIS** to streamline data movement from external sources to the database, ensuring data consistency and reducing manual processing time.
- **Developed** high-impact **Power BI** dashboards using advanced **DAX** measures to visualize metrics from the **Production layer**, driving actionable insights for business reporting.
- **Architected** and managed **OLAP cubes** using **SQL Server Analysis Services (SSAS)**, optimizing multi-dimensional models based on the structured SQL catalogs.
- **Optimized** legacy business reports in **SQL Server Reporting Services (SSRS)** using **MDX** formulas, maintaining continuity for critical financial operations.

D&C Solutions

January 2023 – May 2023

Data Intern

San Nicolás de los Garza, N.L.

- Developed ML projects and early-stage AI solutions using **Python**, **Azure Cognitive Services**, and **Streamlit**. Collaborated on data preparation for model training and designed analysis workflows with **Power BI**.

EDUCATION

Universidad Autónoma de Nuevo León

December, 2022

BS in Computational Science

San Nicolás de los Garza, N.L.

Modules: [Backend Development / Introduction to Machine Learning / Introduction to Deep Learning / Databases].

CERTIFICATIONS

Databricks Certified Data Engineer Associate

Databricks Academy.

May, 2026

Developed and optimized ETL pipelines utilizing PySpark and Delta Lake within Databricks, ensuring efficient data ingestion and transformation.

Databricks Knowledge Badges: Data Ingestion with Lakeflow Connect, Deploy Workloads with Lakeflow Jobs, and Build Data Pipelines with Lakeflow Spark Declarative Pipelines

Databricks Academy.

March, 2026

Databricks Academy Accreditation: Databricks Fundamentals

Databricks Academy.

June, 2025